

A **Computer Vision Engineer** builds systems that can “see” and understand images/videos using AI.

Common work:

- Face recognition
- Object detection
- Thermal camera AI
- CCTV analytics
- Autonomous robots/drones
- Medical image analysis
- Industrial defect detection
- Gesture recognition
- Smart traffic systems

Main technologies:

Programming

- Python
- C++
- CUDA (GPU programming)

AI / Vision Libraries

- OpenCV
- PyTorch
- TensorFlow
- YOLO

Embedded + Edge AI

- ESP32
- Raspberry Pi
- NVIDIA Jetson
- ARM Linux
- TensorRT
- TFLite

Topics to Learn

Beginner

1. Python basics
2. NumPy
3. OpenCV basics
4. Image processing
5. Camera interfacing

Intermediate

1. Object detection
2. Tracking
3. CNN basics
4. YOLO training
5. Dataset creation

Advanced

1. Edge AI deployment
2. CUDA optimization
3. Real-time AI systems
4. Multi-camera systems
5. SLAM
6. Autonomous systems

Projects You Can Build

Beginner Projects

- Motion detector
- Face detector
- QR scanner
- Hand tracking
- Thermal viewer

Intermediate Projects

- AI CCTV
- Vehicle counter
- Helmet detection
- Fire/smoke detection
- Attendance system

Advanced Projects

- Drone vision AI
- Autonomous robot
- Smart traffic AI
- Industrial inspection AI
- Thermal human detection system

Best Hardware for You

Since you already work with ESP32 and thermal cameras:

- ESP32 + camera
- Raspberry Pi 5
- Jetson Nano / Orin Nano
- USB thermal camera
- GPU laptop

Your ASUS TUF laptop with RTX GPU is useful for:

- YOLO training
- AI model testing
- OpenCV projects
- Real-time video processing

Job Roles Related to Computer Vision

- AI Vision Engineer
- Embedded Vision Engineer
- Robotics Vision Engineer
- Edge AI Engineer
- Autonomous Systems Engineer
- Surveillance AI Engineer

Companies Using Computer Vision

- NVIDIA
- Tesla
- OpenAI
- Bosch
- Qualcomm

High Salary Skills

Learn these deeply:

- Linux
- Modern C++
- OpenCV
- YOLO
- CUDA
- TensorRT
- Edge AI deployment
- Embedded systems
- Real-time optimization

A powerful roadmap for you is:

Embedded Systems + Computer Vision + Edge AI + Linux + C++

That combination is highly valuable for robotics, defense tech, drones, AI cameras, and autonomous systems.